Daily Algebra Drill #6

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Comparing coefficients · Substitution $t = \sqrt{x}$ for surd equations · Schur's inequality · Integer-valued polynomials.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

No expansion. Recognise the pattern, write the answer.

1. Factorise: $x^3 + 5x^2 + 6x$.

2. Given $x + \dfrac{1}{x} = \sqrt{5}$, find $x^2 + \dfrac{1}{x^2}$.

3. Without expanding, state the coefficient of xy^2 in $(x + y)^3$.

4. Evaluate: $\dfrac{1}{\sqrt{2}-1} - \dfrac{1}{\sqrt{2}+1}$.

5. The polynomial $f(x) = x^3 + ax + b$ has $x = 2$ as a root and $f'(2) = 0$... but no calculus today! Instead: if $x = 2$ is a *repeated* root, find a and b.

6. Simplify: $\left(\dfrac{2x}{y}\right)^3 \div \dfrac{4x^2}{y^2}$.

7. Given $\alpha\beta = 6$ and $\alpha + \beta = 5$, find $\alpha^2\beta + \alpha\beta^2$.

8. Factorise over $\mathbb{Z}$: $4a^2 - 4ab + b^2$.

9. Show quickly that $2^{10} = 1024$, hence simplify $2^{10} - 1$ as a product of three integer factors greater than 1.

10. Given $f(x) = x^2 - 3x + 2$, find $f(f(0))$.

Section B Manipulation Drills (≤50 s each)

Fractions, structure, identities. Minimum working.

1. Find constants A, B, C such that $x^2 + 3x + 5 \equiv A(x+1)^2 + B(x+1) + C$.

2. The polynomial $p(x) = x^3 + ax^2 + bx + c$ satisfies $p(1) = 0$, $p(-1) = 4$, and $p(2) = 15$. Find a, b, c.

3. Simplify: $\dfrac{x^3-8}{x^2+2x+4}$.

4. Factorise: $a^3+b^3+c^3-3abc$. (*State the identity from memory, then verify by choosing* $a=b=c$.)

5. Given $a+b+c=0$, simplify $\dfrac{a^3+b^3+c^3}{abc}$.

6. Prove that $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ for $n=1,2,3$ by direct verification, then use the formula to find $\displaystyle\sum_{k=1}^{10}(2k-1)^2$.

7. Simplify: $\dfrac{\dfrac{1}{a+b}-\dfrac{1}{a-b}}{\dfrac{1}{a+b}+\dfrac{1}{a-b}}$.

8. Make b the subject: $\dfrac{a-b}{a+b}=\dfrac{c-d}{c+d}$.

9. Given α,β,γ are roots of $x^3-7x+6=0$, find $\alpha^2+\beta^2+\gamma^2$ and $\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2$.

10. Solve: $\dfrac{2}{x^2-1}-\dfrac{1}{x-1}=\dfrac{3}{x+1}$.

Section C Substitution & Structure (≤90 s each)

One clean substitution is worth ten lines of algebra.

1. Solve: $\sqrt{x}-\dfrac{6}{\sqrt{x}}=1$. (*Let* $t=\sqrt{x}$.)

2. Find the minimum value of $x^2-6x+y^2+4y+20$ by completing the square in both x and y. *(after TMUA 2022 Paper 1 Q2)*

3. Show that $a^4+b^4 \geq \dfrac{1}{2}(a+b)^4$ is **false**, then find and prove the correct inequality relating a^4+b^4 to $(a^2+b^2)^2$.

4. Solve: $x^4-2x^3-3x^2+4x+4=0$ given that it has a repeated real root.

5. Given $a,b,c>0$ with $a+b+c=1$, prove that $ab+bc+ca \leq \dfrac{1}{3}$.

6. If $x+y=1$, find the maximum value of x^2y+xy^2.

7. Solve over $\mathbb{R}$: $\sqrt{3x+1}-\sqrt{x+4}=1$.

8. The equation $x^2 + bx + c = 0$ has roots r and s. For what values of b and c are the roots of $x^2 + cx + b = 0$ also r and s?

Section D Challenge (SMC / BMO1 difficulty)

No brute force. Each question rewards one key observation.

1. Schur's Inequality (degree 2 case). Prove that for all non-negative reals a, b, c:

$$a^2(a-b)(a-c) + b^2(b-a)(b-c) + c^2(c-a)(c-b) \geq 0.$$

(*Consider the case $a \geq b \geq c \geq 0$ and group terms strategically.*)

2. Find all polynomials $p(x)$ with real coefficients satisfying $p(x^2) = p(x)^2$ for all real x. *(after BMO1 2015 Q4)*

3. Evaluate:

$$\frac{1^2}{1 \cdot 3} + \frac{2^2}{3 \cdot 5} + \frac{3^2}{5 \cdot 7} + \cdots + \frac{n^2}{(2n-1)(2n+1)}.$$

Express your answer as a closed form in n.

4. A polynomial $p(x)$ of degree 3 satisfies $p(n) = \dfrac{1}{n}$ for $n = 1, 2, 3, 4$. Find $p(5)$.

5. Find all pairs of positive integers (m, n) with $m > n$ such that $m^2 - n^2 = 105$.

"In mathematics the art of proposing a question must be held of higher value than solving it." — Georg Cantor

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Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths